

YAOWU FAN

✉ fanyw5@mail2.sysu.edu.cn · ☎ (+86) 15754954774 · 🌐 github.com/fyw1999 · 🔍 Google Scholar

🎓 EDUCATION

Sun Yat-sen University (SYSU, 985), Guangzhou, China 2024 – Present

Ph.D. in Computer Technology, expected June 2028

School of Computer Science and Engineering; Advisor: Prof. Andy J. Ma

Northwestern Polytechnical University (NWPU, 985), Xi'an, China 2021 – 2024

M.Eng. in Software Engineering

School of Artificial Intelligence, Optics and Electronics (iOPEN); Advisor: Prof. Qi Wang

Inner Mongolia University (IMU, 211), Hohhot, China 2017 – 2021

B.Eng. in Software Engineering; GPA: 3.72/4, Rank: 1/118

📖 PUBLICATIONS AND MANUSCRIPTS

Journal Articles

- [1] **Y. Fan**, J. Wan, and A. J. Ma, “Learning Crowd Scale and Distribution for Weakly Supervised Crowd Counting and Localization,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 35, no. 1, pp. 713–727, 2025, [Paper] [Code].

Conference Proceedings

- [2] **Y. Fan**, J. Wan, T. Han, A. B. Chan, and A. J. Ma, “Video Individual Counting for Moving Drones,” in *IEEE International Conference on Computer Vision (ICCV)*, 2025, pp. 12 284–12 293, **ICCV 2025 Highlight**; [Paper] [Code].
- [3] **Y. Fan**, J. Wan, Y. Yuan, and Q. Wang, “Weakly-Supervised Scene-Specific Crowd Counting Using Real-Synthetic Hybrid Data,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5, [Paper] [Code].

Submitted Manuscripts

- [4] **Y. Fan**, J. Wan, T. Han, A. J. Ma, W. Ouyang, and A. B. Chan, “Video Individual Counting and Tracking from Moving Drones: A Benchmark and Methods,” Submitted to *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2026, [Paper] [Code].
- [5] **Y. Fan**, T. Han, D. Du, A. J. Ma, and J. Wan, “Train the Agent, Not the Expert: Learning to Harness Heterogeneous Experts for Multi-Turn Visual Reasoning,” Submitted to the *Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*, 2026, [Paper].

🔍 RESEARCH EXPERIENCE

Dense Crowd Counting and Tracking from Moving Drones

- Introduced MovingDroneCrowd++, a diverse and challenging dataset specifically designed for video individual counting and tracking in large-scale crowded scenes captured by moving drones under varied heights, angles, and lighting conditions.
- Developed two video individual counting methods and one tracking method for challenging moving drone videos with complex camera motion and dense crowds.
- Achieved significant improvements over existing methods in moving drone scenarios with dense crowds and complex motions, reducing the counting error by 47.4% and improving tracking performance by 64.6%.

Trainable Visual AI Agents for Multi-Turn Visual Reasoning

- Proposed a trainable visual agent, VisHarness, that decouples high-level visual perception, reasoning, and decision-making from low-level visual task execution. It requires only a small amount of training data to learn a generalizable harnessing policy over a carefully designed set of heterogeneous visual experts and can solve various fundamental vision tasks under complex conditions through multi-turn reasoning.

- Experimental results on four benchmarks covering Generalized Referring Expression Segmentation (GRES), Reasoning Segmentation, and referring expression detection and counting for dense small objects show that VisHarness outperforms general-purpose models and achieves comparable or even better performance than specialized models.

Q RESEARCH INTERESTS AND FUTURE DIRECTIONS

- **Computer Vision:** detection, localization, counting, and tracking of dense small objects in open and complex real-world environments.
- **AI Agents:** post-training of multi-turn interactive agents with reinforcement learning, with a focus on visual reasoning and tool/expert coordination.
- **Future research:** developing generalizable visual and embodied agents that connect open-world perception with physical-world interaction, including robotic manipulation and humanoid robots.

♡ HONORS AND AWARDS

National Scholarship, Ministry of Education of China	Jun. 2021
<i>2nd Prize</i> , Software Design and Development Contest for Five Provinces in North China	Jun. 2020

⚙️ SKILLS AND SERVICE

- Programming Languages: Python, C++
- Deep Learning Frameworks: PyTorch, TensorFlow
- LLM Training Frameworks: veRL, ms-swift
- Languages: CET-6: 475
- Academic Service: Reviewer for TCSVT, ICCV, and PRCV